Early Time Series Classification Using TCN-Transformer

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***Abstract*—Early time series classification is of great significance for time-sensitive applications such as fault detection and earthquake prediction. This task aims to classify time series with the least timestamps at desired accuracy. Recent deep learning methods usually used the Recurrent Neural Networks (RNNs) as the classification backbone and the exiting subnet for early quitting. However, the RNNs suffer from the 'forgetting' defect and insufficient local feature extraction. Besides, the balance between earliness and accuracy is not fully considered. In this paper, a framework named TCN-Transformer is proposed. To overcome the defects of RNNs, we combined Temporal Convolutional Network and Transformer to extract both local and global features. Then, a loss function is designed to ensure the classification performance, while focusing more on earlier features. The experimental results on ten univariate datasets.**

Keywords—early classification, Transformer, deep learning

I. Introduction

With the development of sensor techniques, time series classification has been widely studied in many fields, e.g. human action recognition, biomedicine, and smart home [1-3]. For some time-sensitive application fields such as gas leak detection and fault diagnosis, Early Time Series Classification (ETSC) has attracted extensive attention[4]. The task of ETSC is to identify categories of observable time series as accurately and quickly as possible[5].

In the last few years, deep learning methods have attracted extensive attention in ETSC due to their effective feature representation. Therefore, this paper focuses on deep learning methods in ETSC. Since the classification accuracy generally decreases with the information reduction, earliness and accuracy are two conflicting targets. To balance earliness and accuracy, some deep learning methods first trained a classifier, and then set a threshold or rules to exit. However, the two-stage approach completely separates the exit process from the classification process and cannot extract effective early features. To handle this problem, some one-stage methods used the end-to-end framework to jointly optimize the classification and early exiting subnets. By reducing the reward for correct classification over time, these methods enable the model to focus on early features. Unfortunately, this reduction is linear and does not meet real-world needs. Besides, in the real world, the length of input data increases over time for ETSC. To adapt to the length variation of the input data, many methods generally use Recurrent Neural Networks (RNNs) as their classification backbone. However, the RNNs are difficult to construct long-term correlations due to the 'forgetting' defect caused by the recursive structure. Moreover, the local feature extraction ability of RNNs is insufficient.

To tackle these problems, this paper proposes a hybrid model named TCN-Transformer and a loss function based on time decay. First, to effectively balance accuracy and earliness, a loss function based on time decay is designed. By changing the decay rate of classification rewards at different times, the loss function can better extract features. Then, to overcome the defects of RNNs, TCN-Transformer integrates the Temporal Convolutional Network (TCN) and Transformer. In this way, TCN-Transformer better extracts both local and global features.

(1) To overcome the defects of RNNs, we combined TCN and Transformer to effectively extract local and global features.

(2) To effectively balance accuracy and earliness, we design a loss function based on time decay that focuses more on earlier sequences.

(3) The proposed method achieves superior performance on ten commonly used datasets.

II. Method

A. Overview

To solve the forgetting defect of RNNs and the weak ability of local feature extraction, we propose a TCN-Transformer to extract local features and global features. And a loss function is designed to extract earlier features. The overall architecture of the proposed model is presented in Fig. 1. As shown, it consists of the TCN-Transformer module, classification subnet, and the early-exit subnet. First, the time series data is input into the TCN-Transformer module to extract features. Then the extracted features are fed into the classification subnet and the early exit subnet to obtain the classification probability Pt and the exit probability βt, respectively. In the test phase, when βt is less than 0.5, the classification probability Pt is not available at this time, and then data continues to be fed into the network over time. When βt is greater than 0.5, we consider the classification result at this time as the final classification result and stop continuing to input data.

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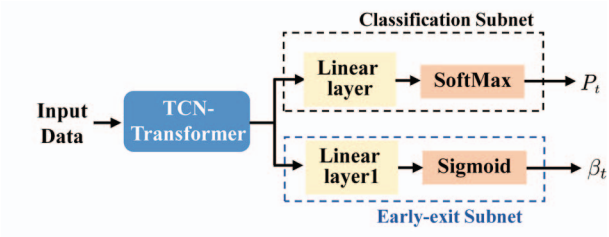

Fig. 1. The overall architecture of the proposed model. Pt is the classification probability at time step t. βt is the exit probability at time step t.

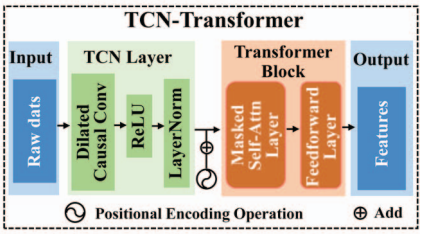

Fig. 2. The architecture of the TCN-Transformer module

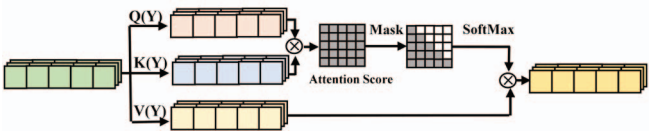

Fig. 3. The architecture of the masked self-attention layer

B. The Architecture of TCN-Transformer

To better fuse local features and global features, we propose a TCN-Transformer network composed of the TCN layer and the Transformer blocks. The architecture of the proposed TCN-Transformer is presented in Fig. 2. First, the input data is fed into the TCN layer to efficiently extract local features. To make the output at time step j is only convolved with features at time step j or earlier and increase the receptive field of the network, dilated causal convolution is used at each TCN layer. LayerNorm is added to the TCN layer for normalization. Moreover, the positional encoding is added for preserving the temporal order. Second, the local features extracted by the TCN layer are fed to the Transformer block to extract global features. Each block consists of a self-attention layer and a feed forward layer. For conventional self-attention mechanisms, the features at each time step are associated with the features at all time steps in the sequence data. A masked self-attention layer to make the features of each time step only associated with the features of the observed data at the current moment, we design. The masked self-attention layer is illustrated in (1) and Fig.3.

$$Atten(Q,K,V) = SoftMax\left(Mask\left(\frac{QK^T}{\sqrt{d_k}}\right)\right) * V \quad (1)$$

In Fig.3 and (1), Q (query), K (key), and V (value) in Fig.3 are the learned linear transformation of the input, and Y in Fig.3 refers to the extracted local features. The attention is obtained by computing the dot products of the query with all keys. To not leak the information after the current moment, the upper triangle of the attention score is set to negative infinity. The upper triangle of the attention score is made 0 by applying the softmax function. This process of further processing the attention score is named Mask. Finally, the product of the processed attention score and value is fed into the feed-forward layer. The feed-forward layer is a Multilayer Perceptron applied to each timestamp.

C. Loss Function

To extract earlier features, we design a loss function defined as

$$L(\theta) = \frac{1}{N}\sum_{i=1}^{N}\sum_{t=1}^{T}\delta_t^i * [-a * \log P_t^i + (1-a) * (1-(t/T)^2) * P_t^i], \quad (2)$$

where θ refers to all the parameters of the model, P_t^i refers to the classification probability of the correct class of the sample i at time t, parameter a controls the balance of earliness and accuracy, N is the number of dataset samples and T is the full time series data length. In(2), δ_t^i defined as

$$\delta_t^i = \beta_t^i \prod_{j=1}^{t-1}(1-\beta_t^i), \quad (3)$$

where β_t^i refers to the exit probability of sample i at time t. δ_t^i means the exiting probability at time t under the condition that the classifier cannot exit classifying before time t. And $\log P_t^i$ and $1-(t/T)^2 * P_t^i$ in (2) represent the loss of misclassification and the reward of correct classification at that moment, respectively. The correctly classified reward decays over time. At earlier timestamps, the reward is larger and decays slower, and at later timestamps, the reward is smaller and decays faster. This encourages the model to focus on earlier features and exit classification at an earlier timestamp.

III. Experiment & Discussion

A. Experimental Setup Dataset

1) Dataset Description

Ten UCR datasets are used for performance evaluation. The ten univariate datasets are derived from the commonly used 45 UCR datasets. The principle of our selection of datasets is that the number of training sets exceeds 300. In the end, we removed some highly similar datasets from it.

2) Experimental Settings

All experiments were performed using Pytorch 1.9.0 on a computer with an Nvidia RTX 3080 GPU. An Adam optimizer with an initial learning rate of 0.001 is used to update all parameters of the model. The epoch is set to 150.

We choose the model with the lowest loss on the training dataset for performance evaluation. The batch size is determined by the number of samples in the dataset.

3) Evaluation Rule

In this paper, to comprehensively measure the early classification performance, the harmonic mean (HM) of earliness and accuracy as the evaluation criteria. The HM is defined as

$$H(M) = \frac{2*accuracy*(1-earliness)}{accuracy+(1-earliness)}. \quad (4)$$

B. Comparisons of Different Methods

We compare our method with several deep learning methods, including EARLIEST, ETMD, and ECLN. The results of the comparison are shown in Table I.

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TABLE I. CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS

Datasets	Methods			
	ECLN	ETMD	EARLIEST	TCN-Transformer
ChlorineCon	69.44	69.35	**69.85**	69.35
CricketX	16.33	26.02	23.84	**56.20**
FaceAll	16.84	7.65	45.22	**70.88**
MedicalImages	52.08	4.86	67.81	**78.00**
NonInvThorax2	80.31	0.15	23.85	**86.11**
StarLightCurves	29.57	81.17	**90.51**	75.22
SyntheticControl	22.83	29.59	68.04	**75.47**
TwoPatterns	17.42	42.84	38.21	**59.71**
UWaveZ	41.03	27.87	44.85	**62.58**
Wafer	86.01	94.00	93.92	**96.77**

From Table I, the proposed method achieves the best early classification performance on most datasets compared with its competitors.

C. Ablation Studies

To illustrate the effectiveness of the TCN-Transformer model and the proposed loss functions, the ablation experiments are divided into the following two parts.

1) TCN-Transformer

To illustrate the proposed TCN-Transformer model can overcome the defects of RNNs, we replace the TCN-Transformer with Long Short-Term Memory (LSTM). The result of the ablation experiment is present in Table II. The result of the experiment shows that our proposed TCN-Transformer can achieve better early classification performance on most datasets.

TABLE II. CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS

Datasets	LSTM	TCN-Transformer
ChlorineCon	69.21	**69.35**
CricketX	21.40	**56.20**
FaceAll	**74.96**	**70.88**
MedicalImages	53.00	**78.00**
NonInvThorax2	7.25	**86.11**
StarLightCurves	14.23	75.22
SyntheticControl	35.64	**75.47**
TwoPatterns	15.78	**59.71**
UWaveZ	36.71	**62.58**
Wafer	89.98	**96.77**

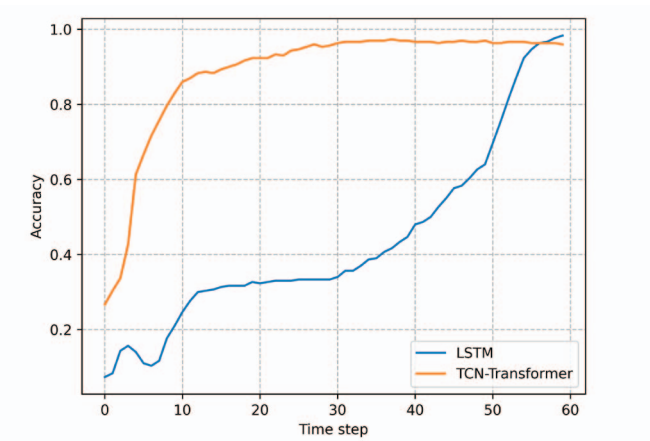

Fig. 4. The accuracy curve of TCN-Transformer and LSTM models at different times on SyntheticControl dataset.

To further demonstrate that the TCN-Transformer model can better extract features and improve the classification performance of time series, the accuracy curve of TCN-Transformer and LSTM models at different times on the SyntheticControl dataset is plotted in Fig.4. From Fig.4, we can find that compared with LSTM, TCN-Transformer can classify more accurately at any time.

2) Loss function

To illustrate the effectiveness of the proposed loss function, we take the loss function mentioned in ECLN as the baseline, and replace our loss function with the baseline. The result of this ablation experiment is shown in Table III. From Table III, it can be found that the proposed loss function significantly improves the early classification performance.

TABLE III. CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS

Datasets	baseline	Proposed loss function
ChlorineCon	**69.71**	69.35
CricketX	41.62	**56.20**
FaceAll	20.99	**70.88**
MedicalImages	76.83	**78.00**
NonInvThorax2	60.39	**86.11**
StarLightCurves	**75.69**	75.22
SyntheticControl	22.98	**75.47**
TwoPatterns	33.84	**59.71**
UWaveZ	36.15	**62.58**
Wafer	96.71	**96.77**

To demonstrate that the proposed loss function can make the classifier pay more attention to the early features and further improve the earliness, the accuracy curve of the TCN-Transformer model trained through the proposed loss function and the baseline is plotted in Fig5. From Fig5, we can find that the proposed loss function improves the early accuracy of the classifier, thereby improving the early classification performance of time series.

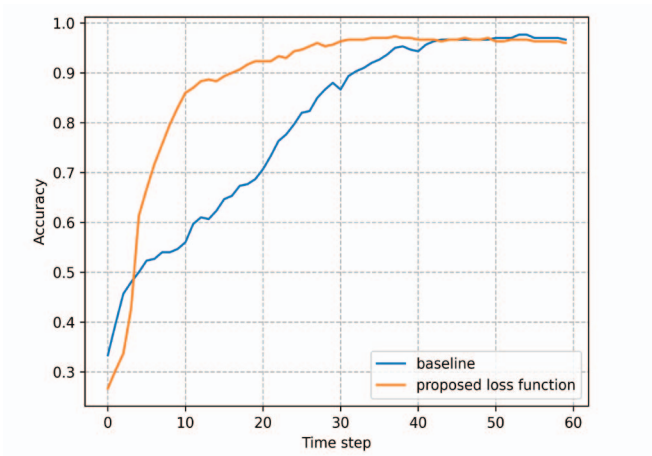

Fig. 5. The accuracy curve of TCN- Transformer trained through proposed loss function and baseline at different times on SyntheticControl dataset.

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IV. Conclusion

This paper proposes an ETSC framework named TCN-Transformer. It contains a TCN-Transformer network and a loss function. TCN-Transformer outperforms RNNs in capturing local features and modeling long-term dependencies. And the proposed loss function better balances earliness and accuracy by paying more attention to early features. Experimental results on ten commonly used datasets demonstrate the superior performance of our proposed framework. However, it remains to be explored how to dynamically extract early features according to data characteristics. In future work, this problem will be further researched.

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